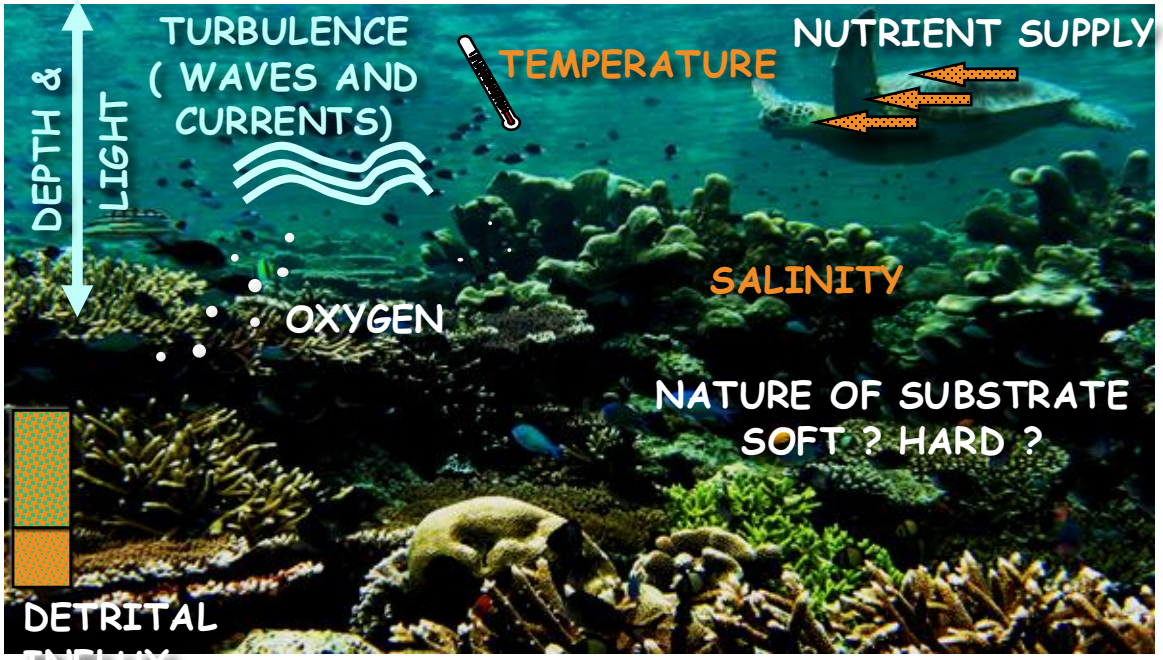
Carbonate factory & carbonate platforms



Review
Controls on carbonate deposition

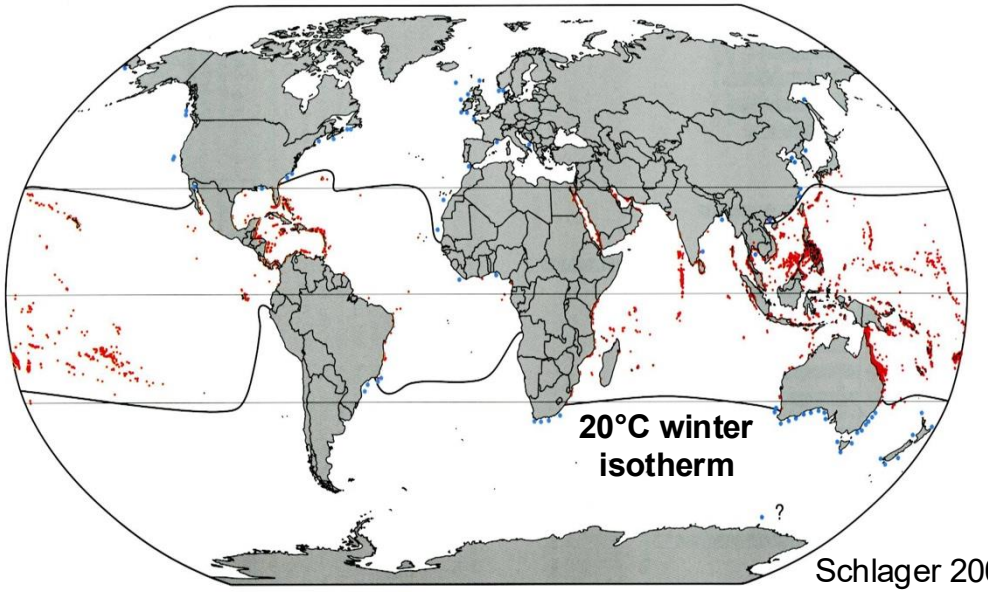


- Essentially limestone ($CaCO_3$) and dolomite ($CaMg(CO_3)_2$)
- Bio-chemical precipitation in sub-tropical/tropical shallow-water areas
- Rocks forming mostly in-situ in marine or in lacustrine settings

$$Ca^{2+} + 2HCO_3^- \leftrightarrow CaCO_3 + H_2O + CO_2$$

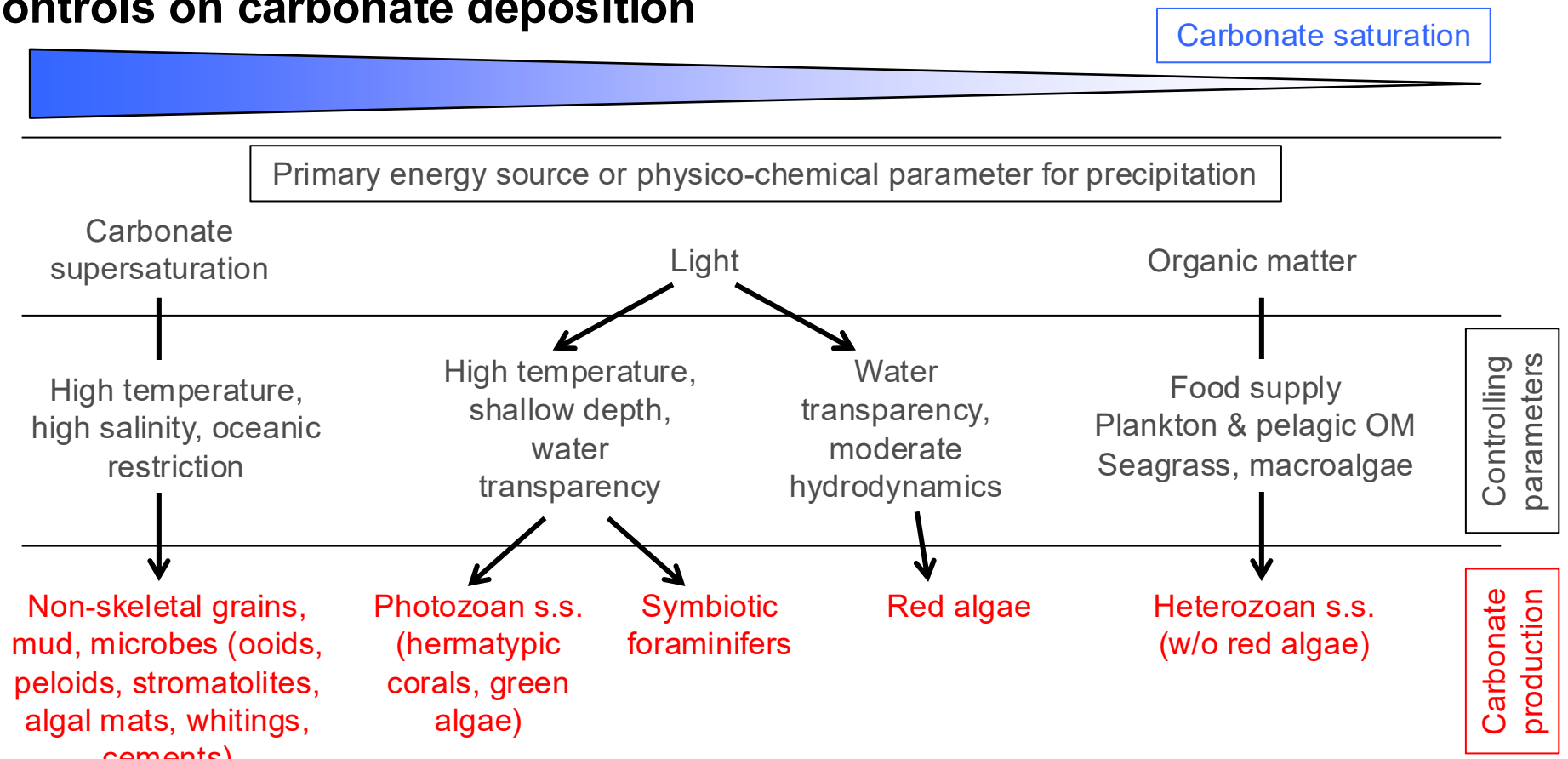
Control of CO_2 on carbonate precipitation/dissolution

Distribution modern scleractinian corals (red)



Schlager 2005

Carbonate Factory
Controls on carbonate deposition



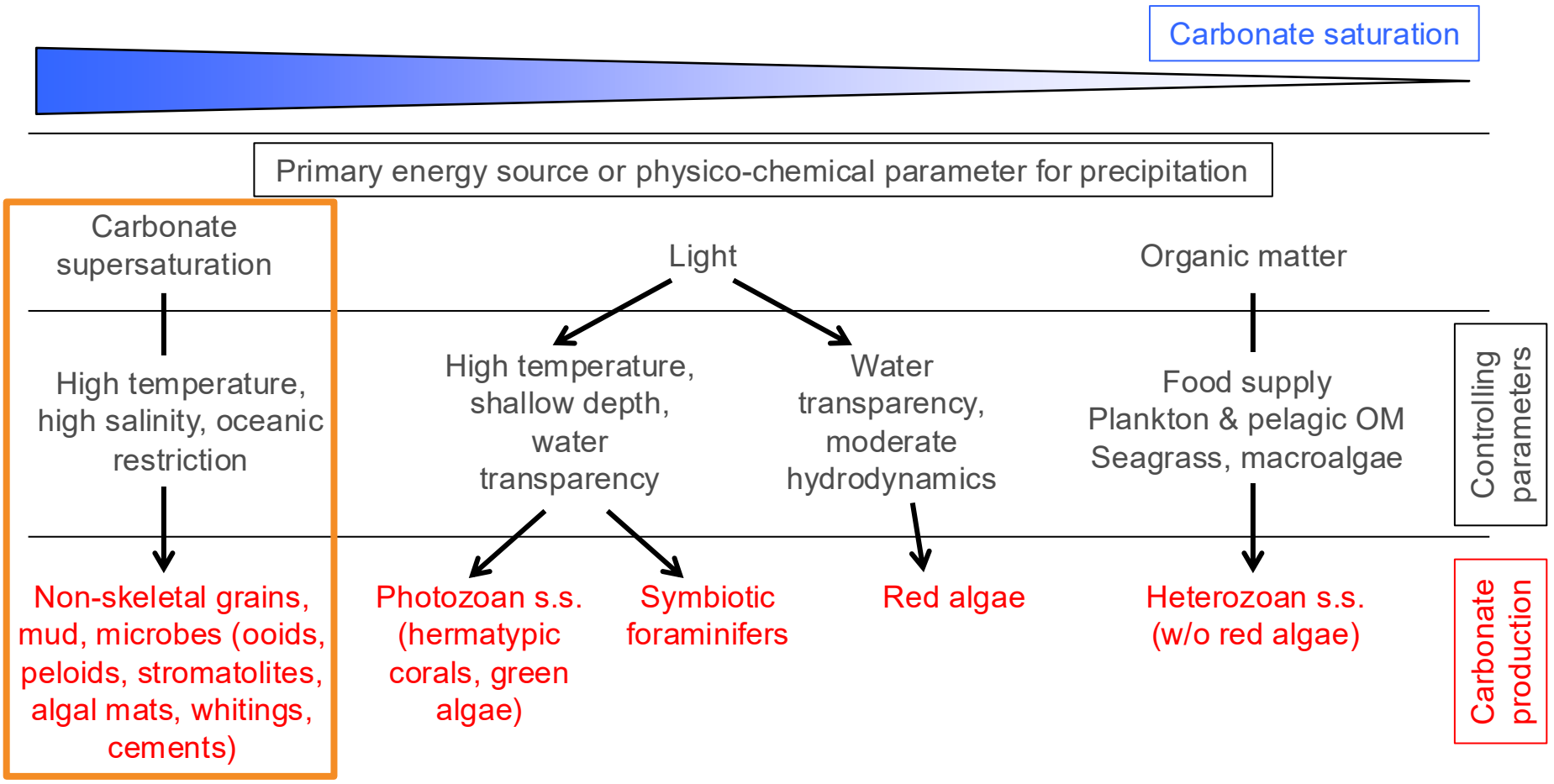
Michel, 2018

Biologically controlled

Variable degrees of biologically influenced/induced

Abiotic

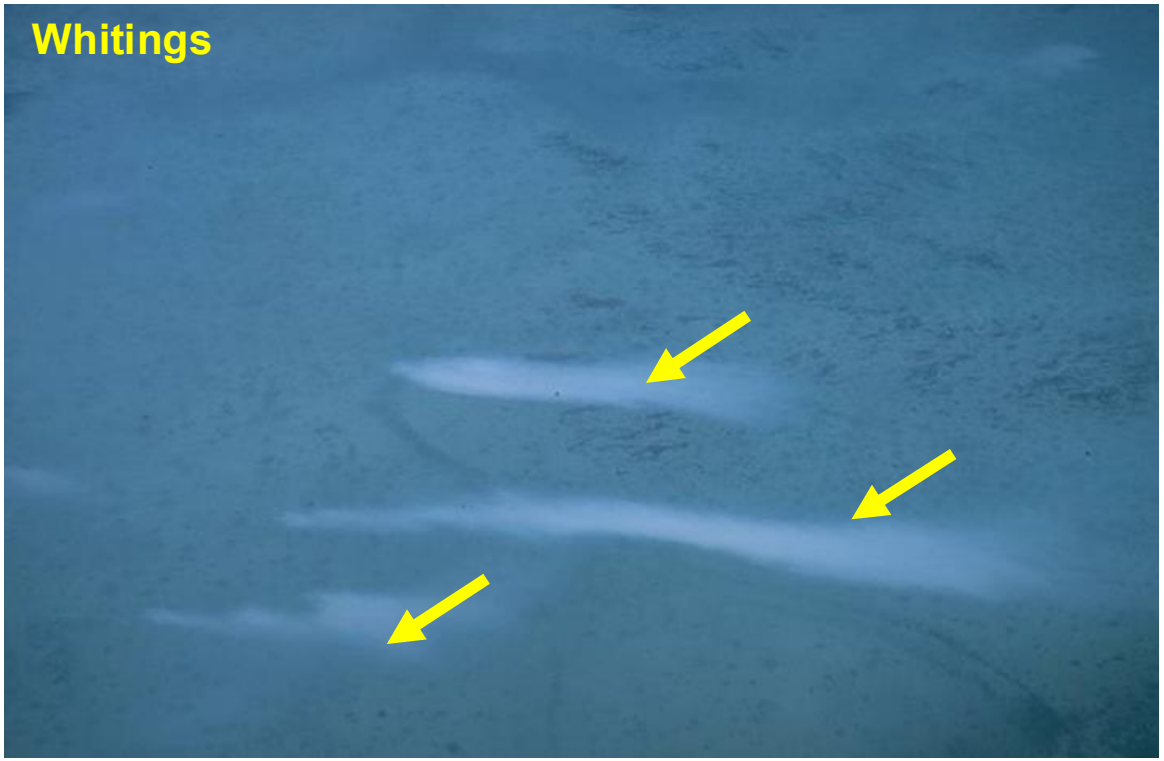
Carbonate Factory
Controls on carbonate deposition



Michel, 2018

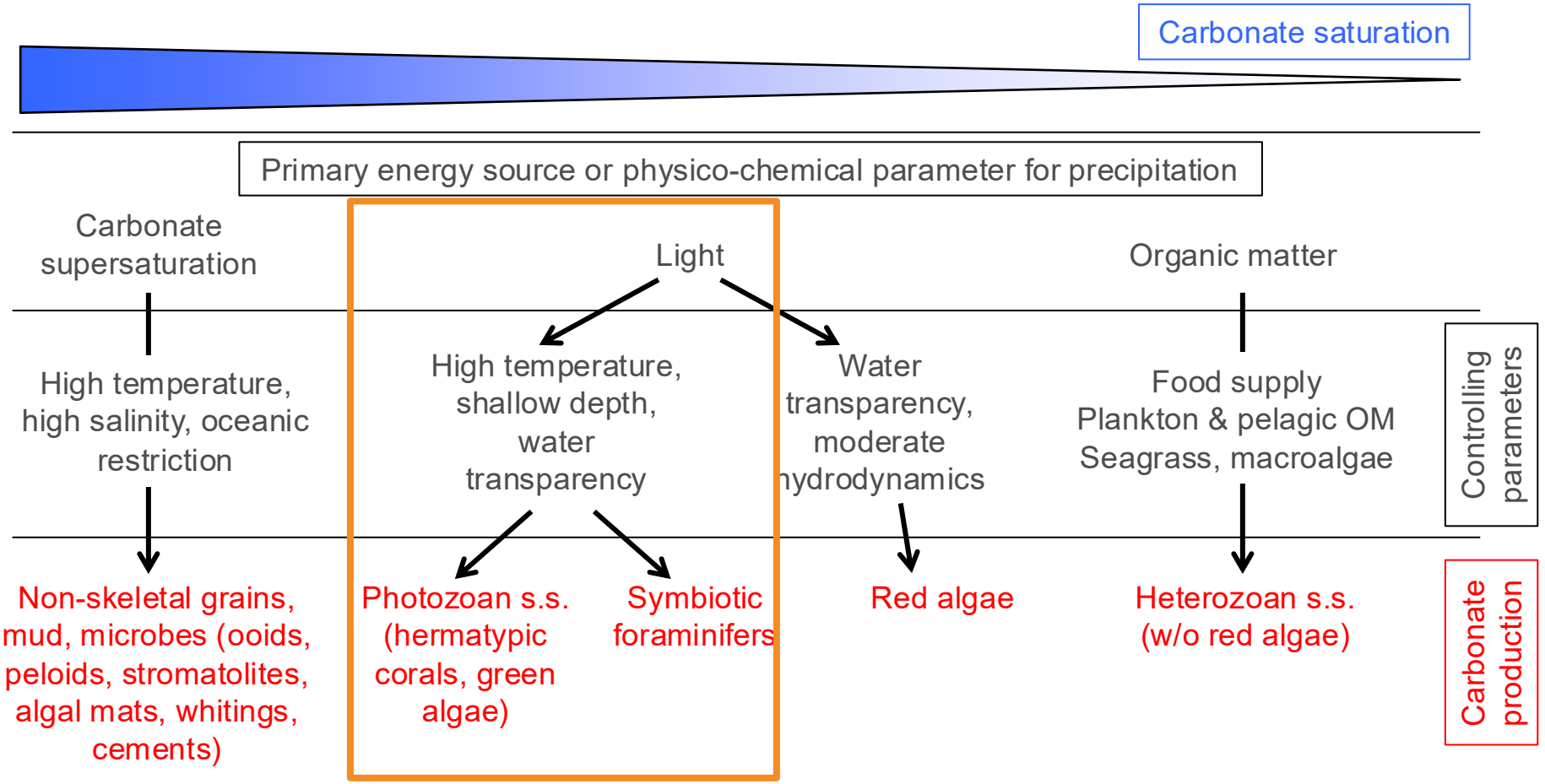
Carbonate Factory
Chemical precipitation

- Modern day precipitation of aragonite in regions of elevated salinity (whitings)
- Ooid precipitation in high energy settings
- Chemical, abiotic precipitation also common in lacustrine settings
- Biological mediation possible: biologically influenced or induced



Near Hicks Cay, Belize, Central America
"Whitings" in the northern Belize shelf lagoon, elongated in a general east to west direction. The uppermost "whiting" is on the order of 300m long.
E. G. Purdy / SEPM

Carbonate Factory
Controls on carbonate deposition



Michel, 2018

Carbonate Factory
In-situ growth

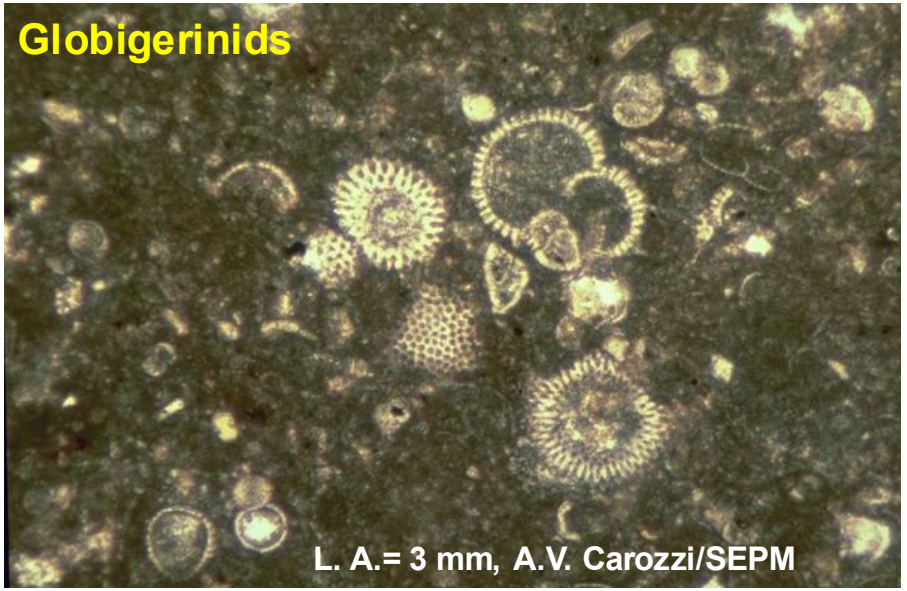
- Carbonate build-ups form by *in-situ* growth of skeletons, many of autotrophic organisms:
  - Corals
  - Algae
  - Stromatoporoids
- Growth of benthic algae creates high volumes of carbonate mud

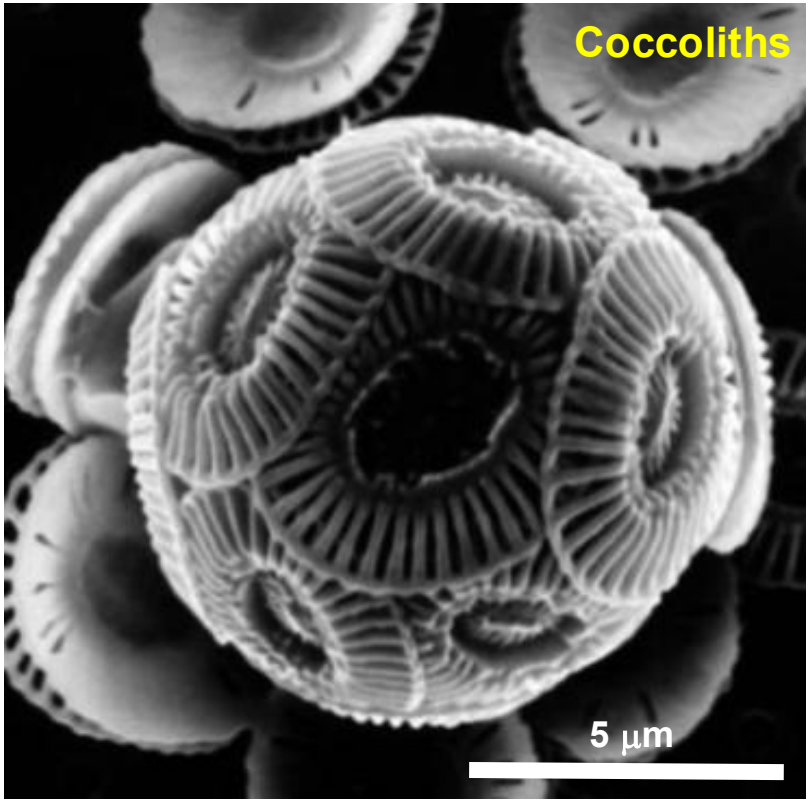




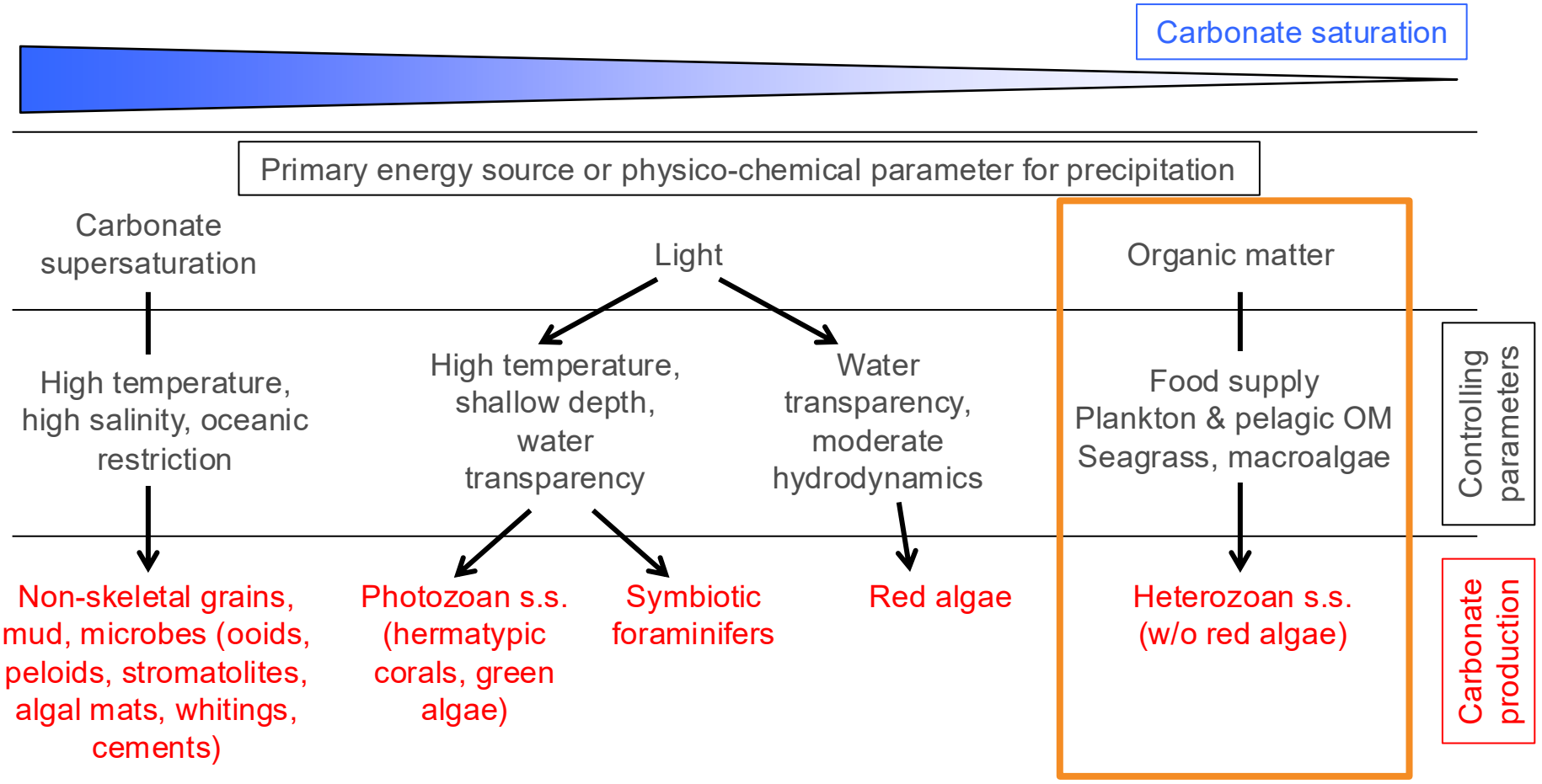
Carbonate Factory
Calcareous plankton

- 80% of present day carbonate deposition is as carbonate ooze, derived from coccoliths and planktonic forams
- Much of this is driven by photosynthesis of autotrophic organisms





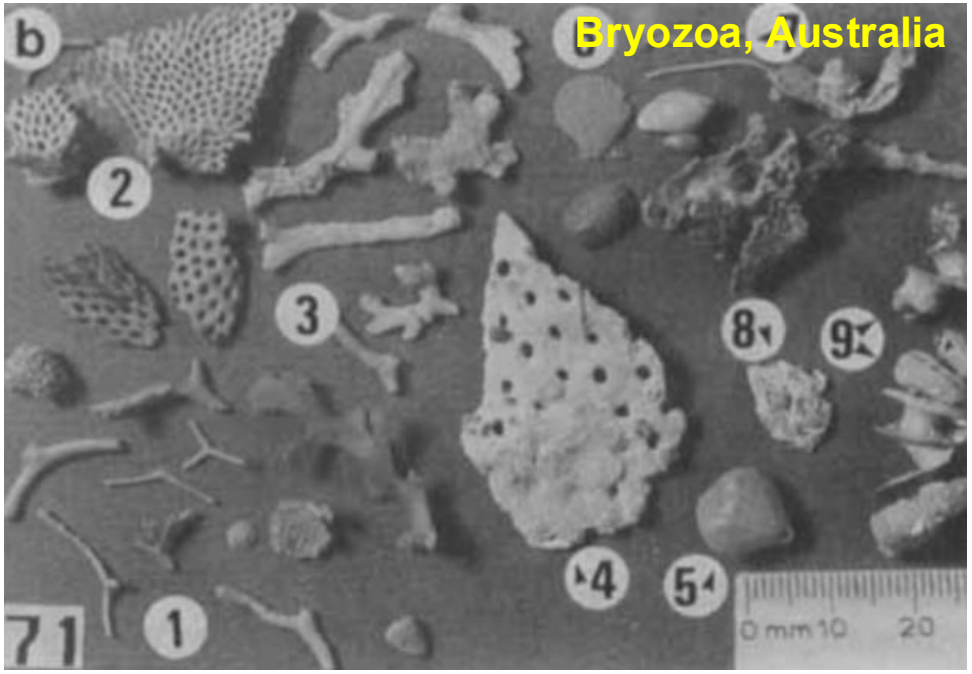
Carbonate Factory
Controls on carbonate deposition



Michel, 2018

Carbonate Factory
Shell accumulation

- Heterotrophic benthic/planktonic organisms contribute to carbonate deposition in the shallow to deep marine environments
- Includes construction of in-situ buildups, e.g. by:
  - Bryozoa
  - Crinoids
  - Bivalves

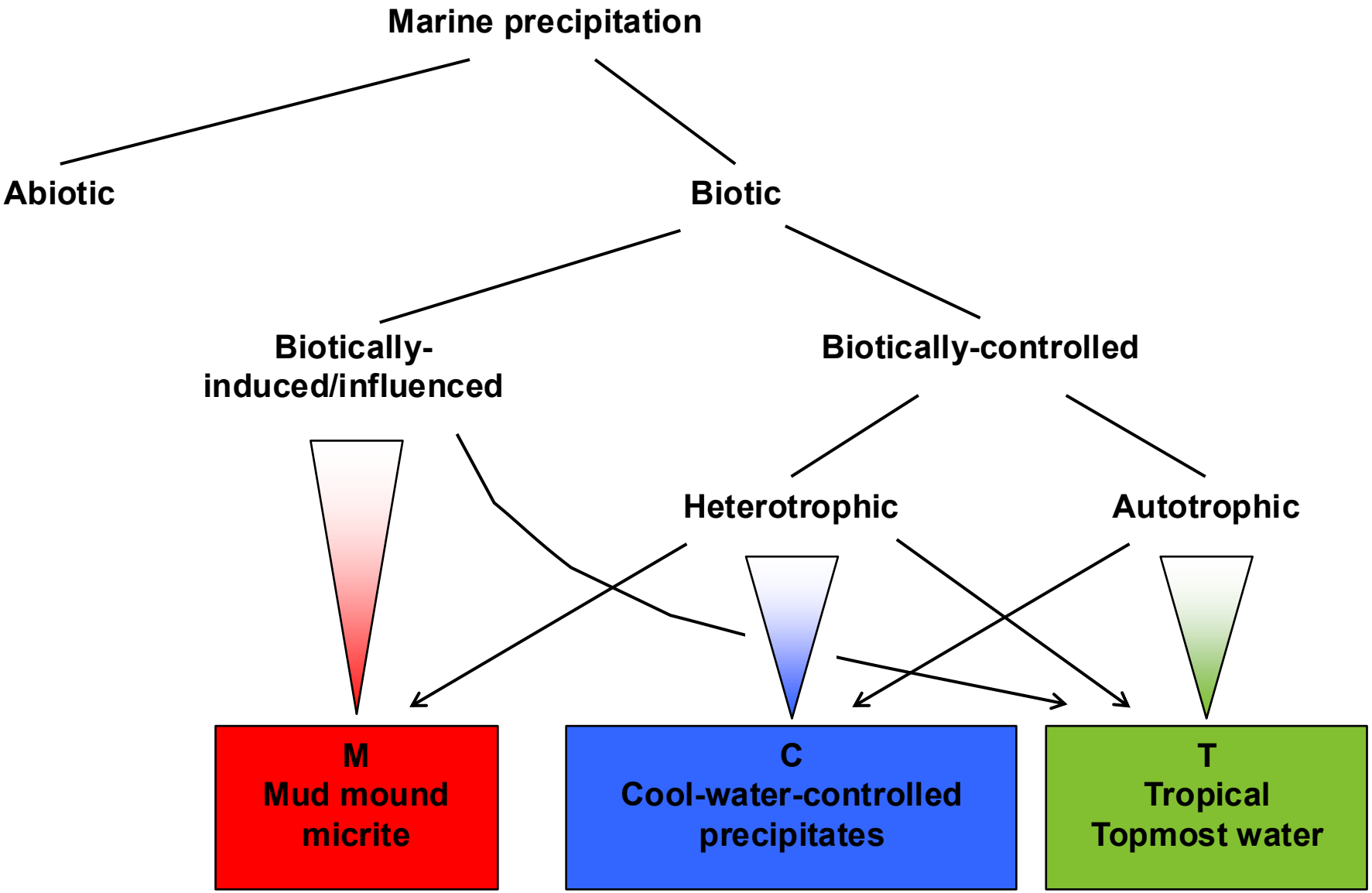


James (1992)



Scholle (1983)

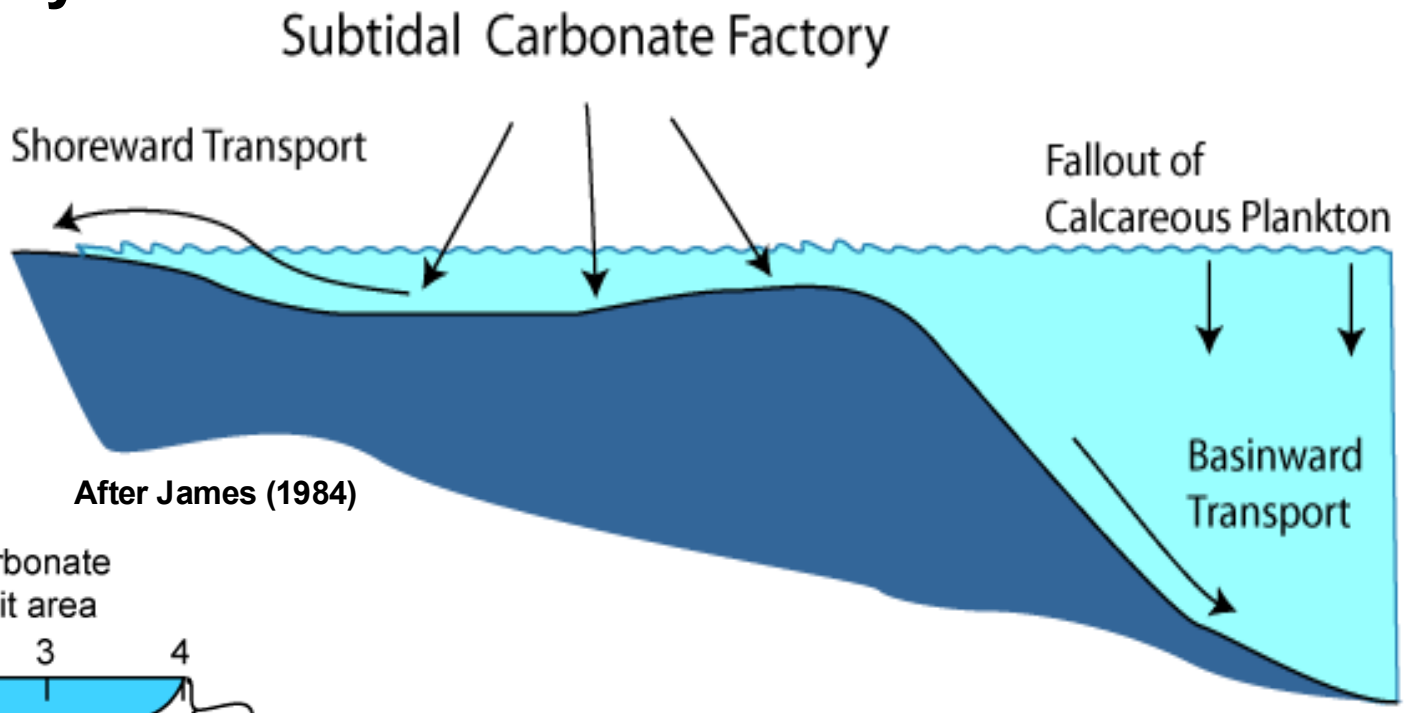
Carbonate Factory
Production modes



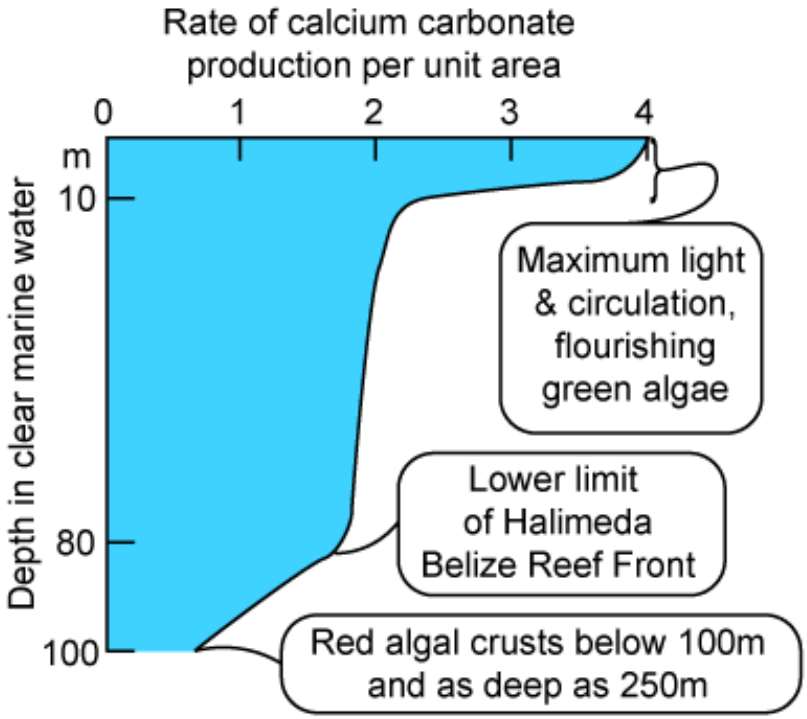
Schlager, 2005

Carbonate Factory

T
Tropical
Topmost water



After James (1984)

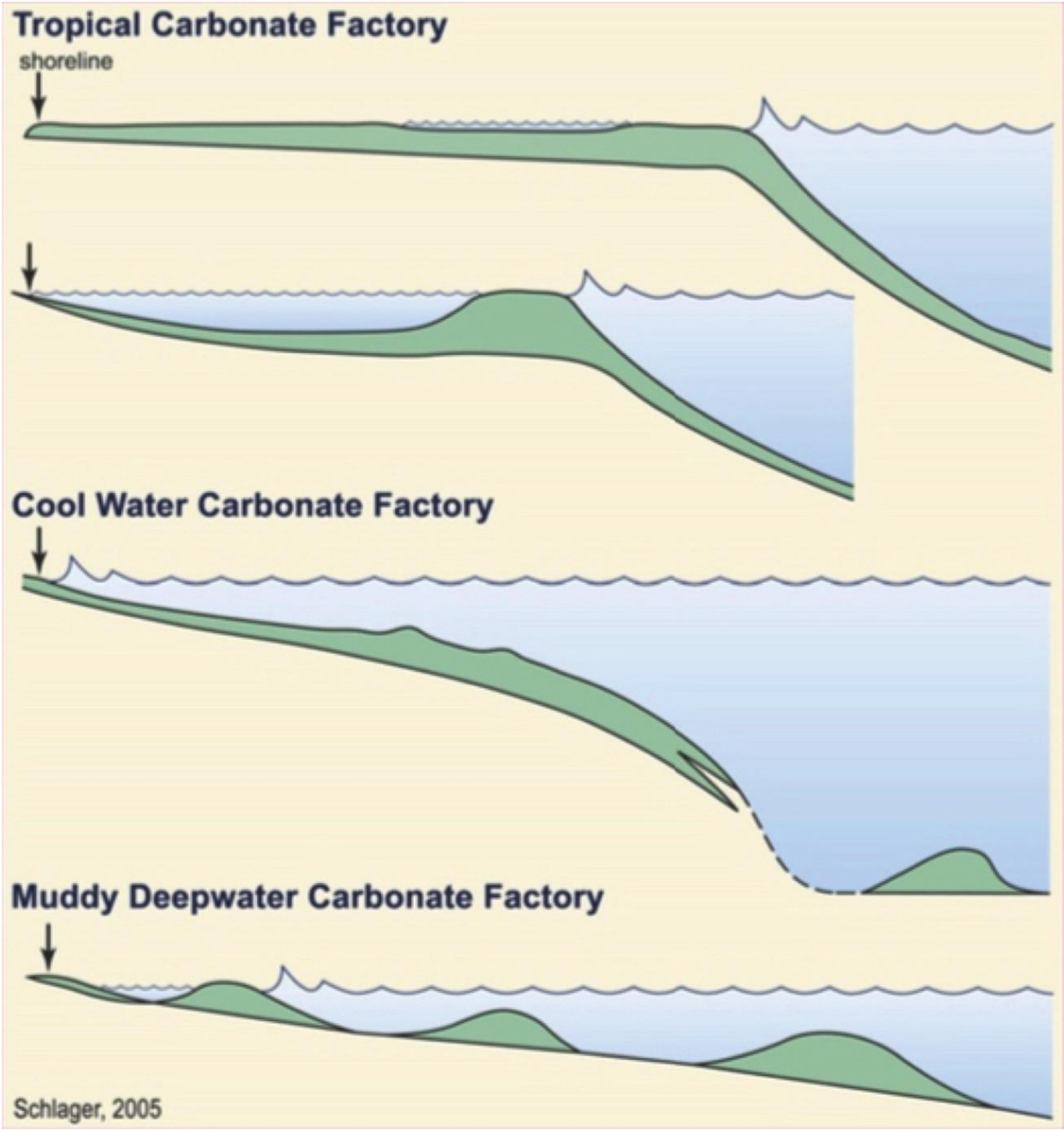


Carbonate Platforms
Production modes

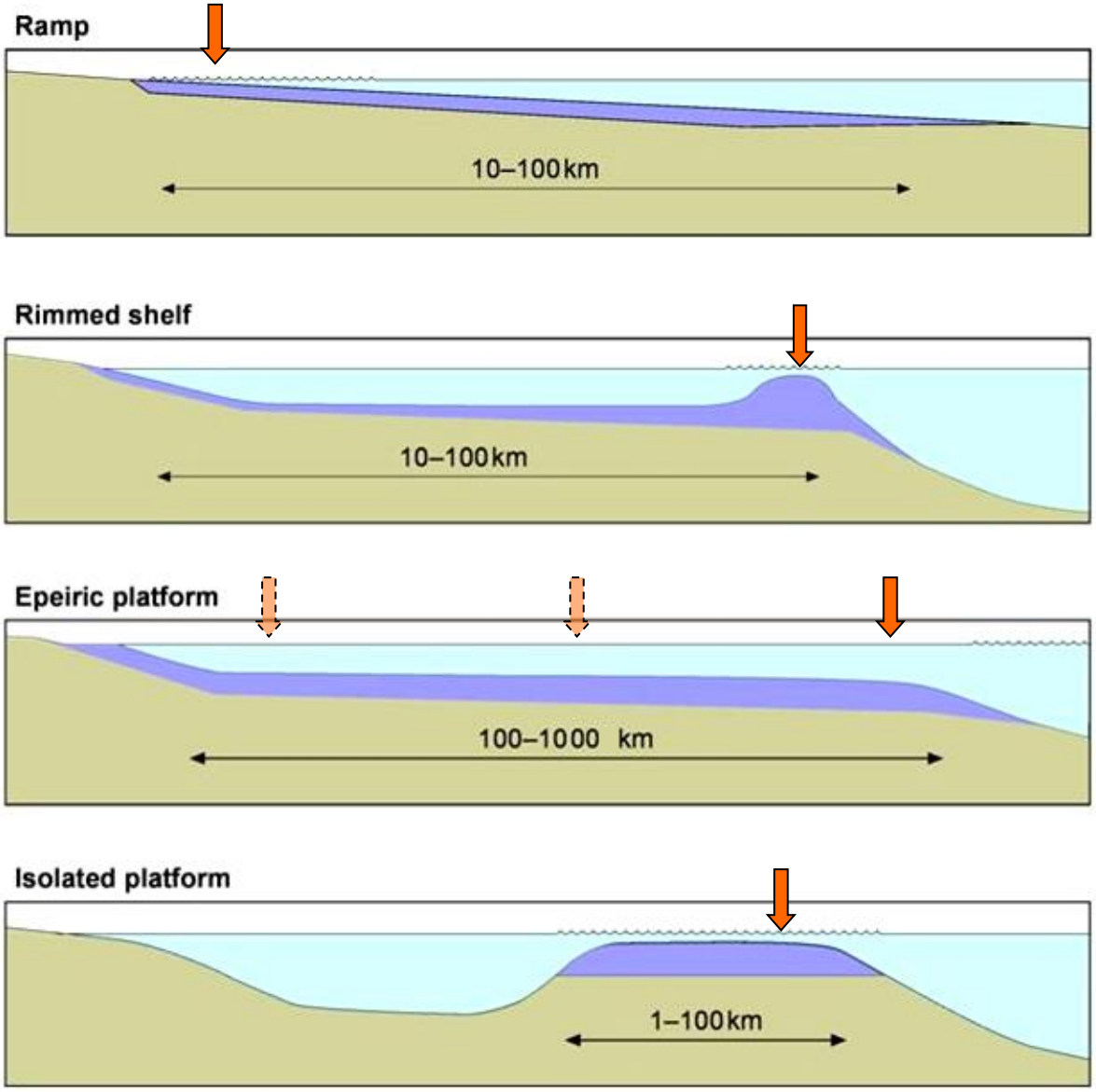
T
Tropical
Topmost water

C
Cool-water-controlled precipitates

M
Mud mound micrite



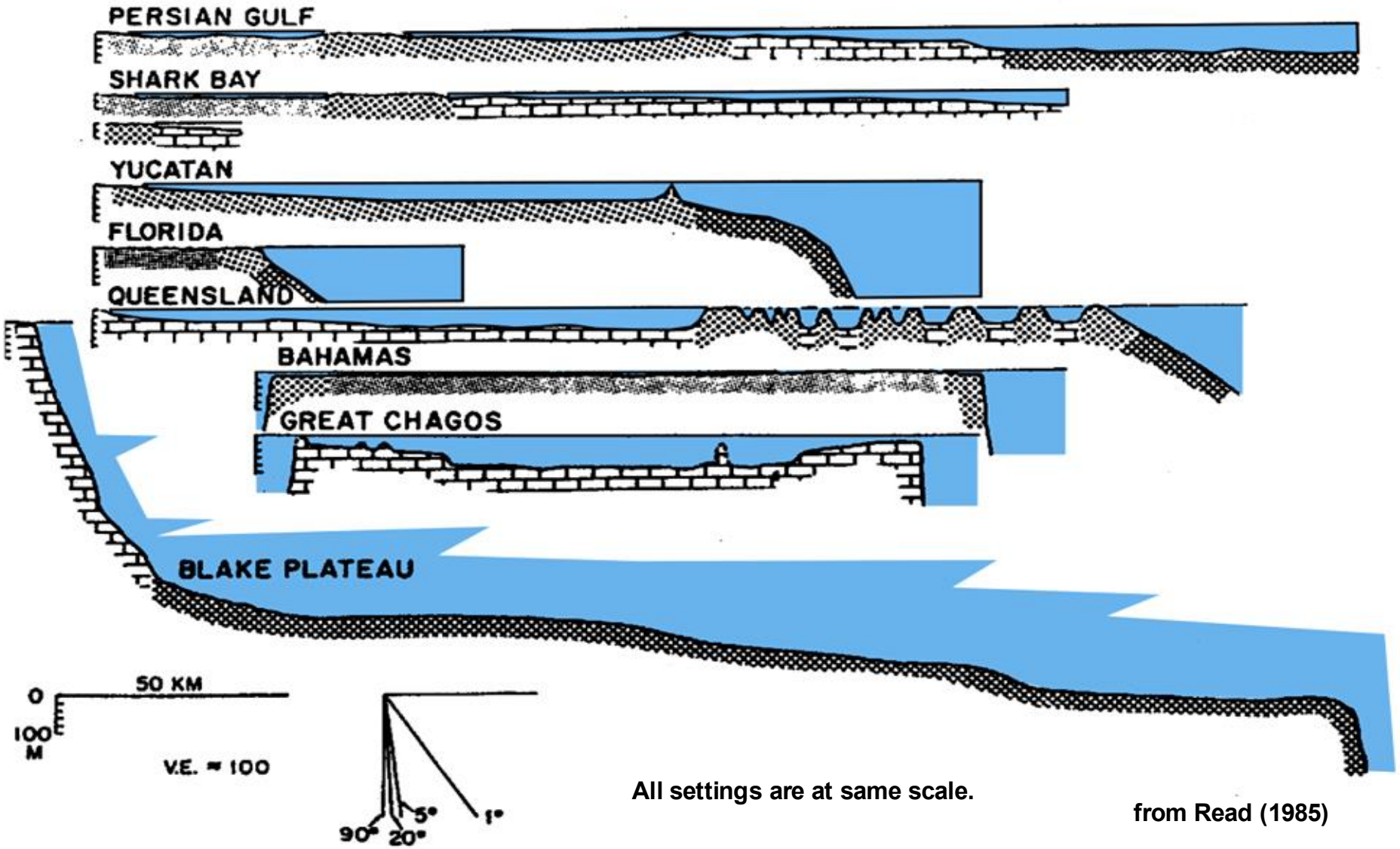
Carbonate Platforms
Classification



Shallowest areas with waves & currents & photic energy -> main carbonate production; best locations for emersion/karstification

Nichols (2009): "Sedimentology & stratigraphy"

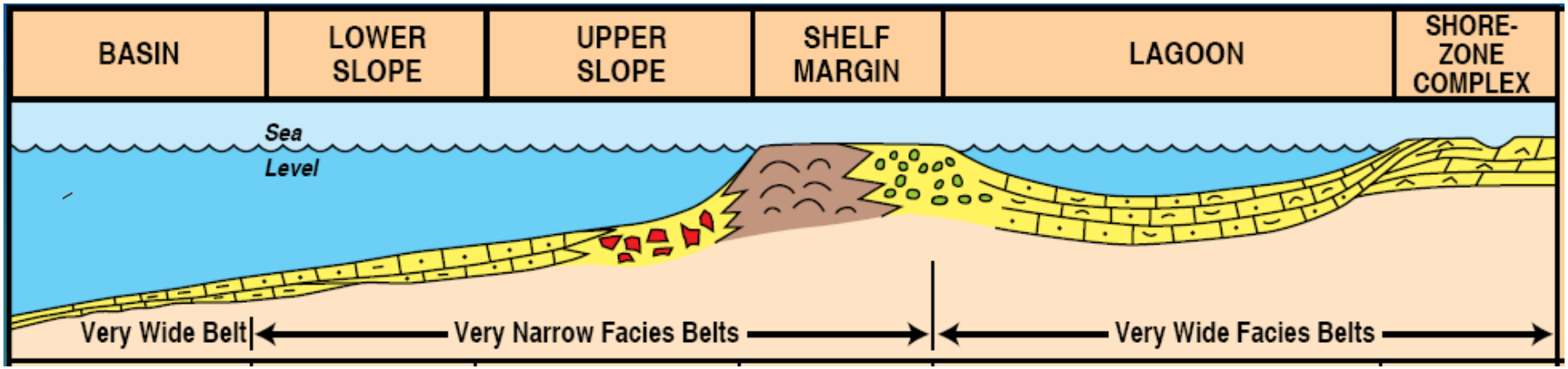
Carbonate Platforms
Classification



All settings are at same scale.

from Read (1985)

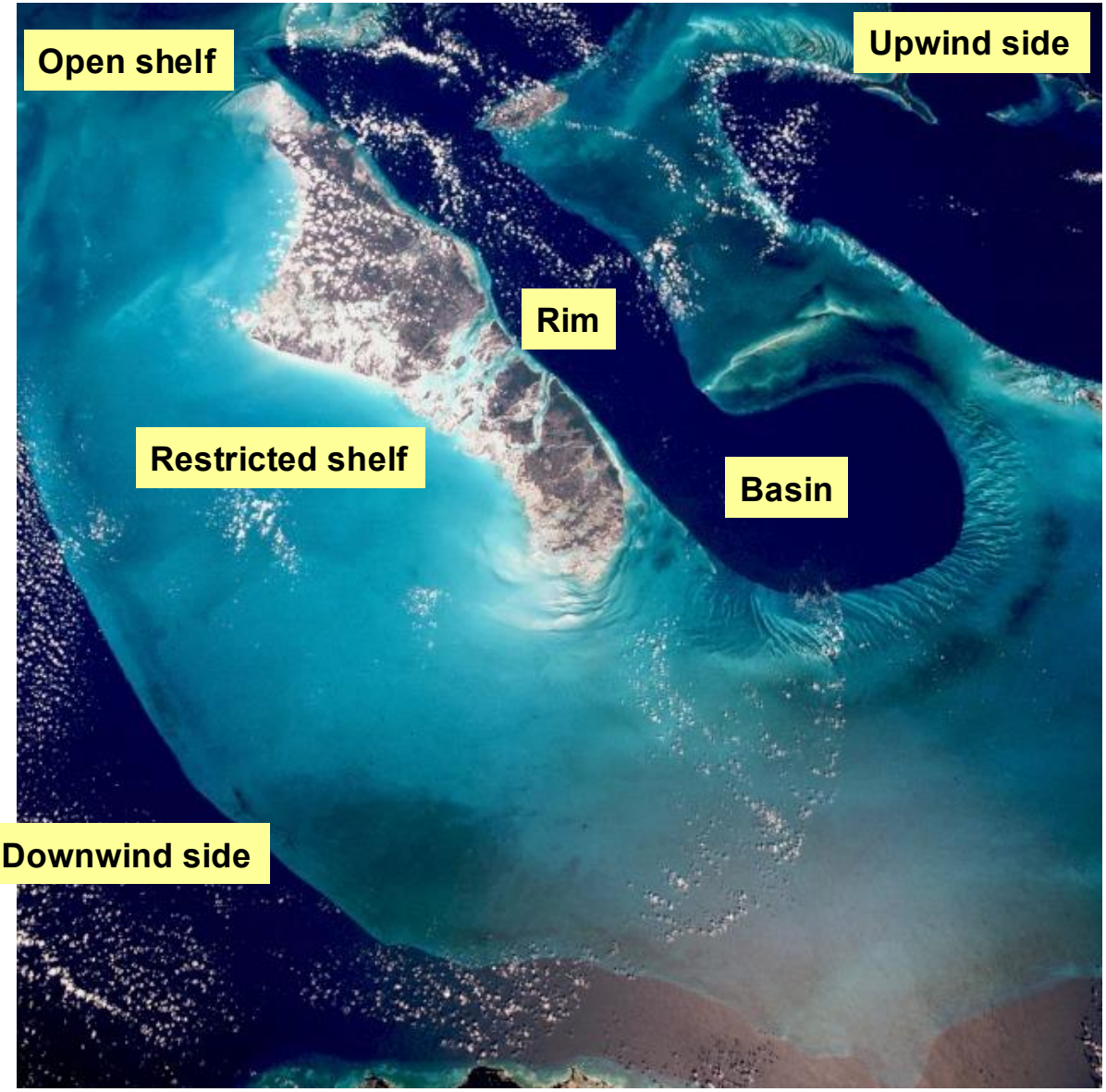
Carbonate Platforms
Rimmed shelf



Moore, 2001

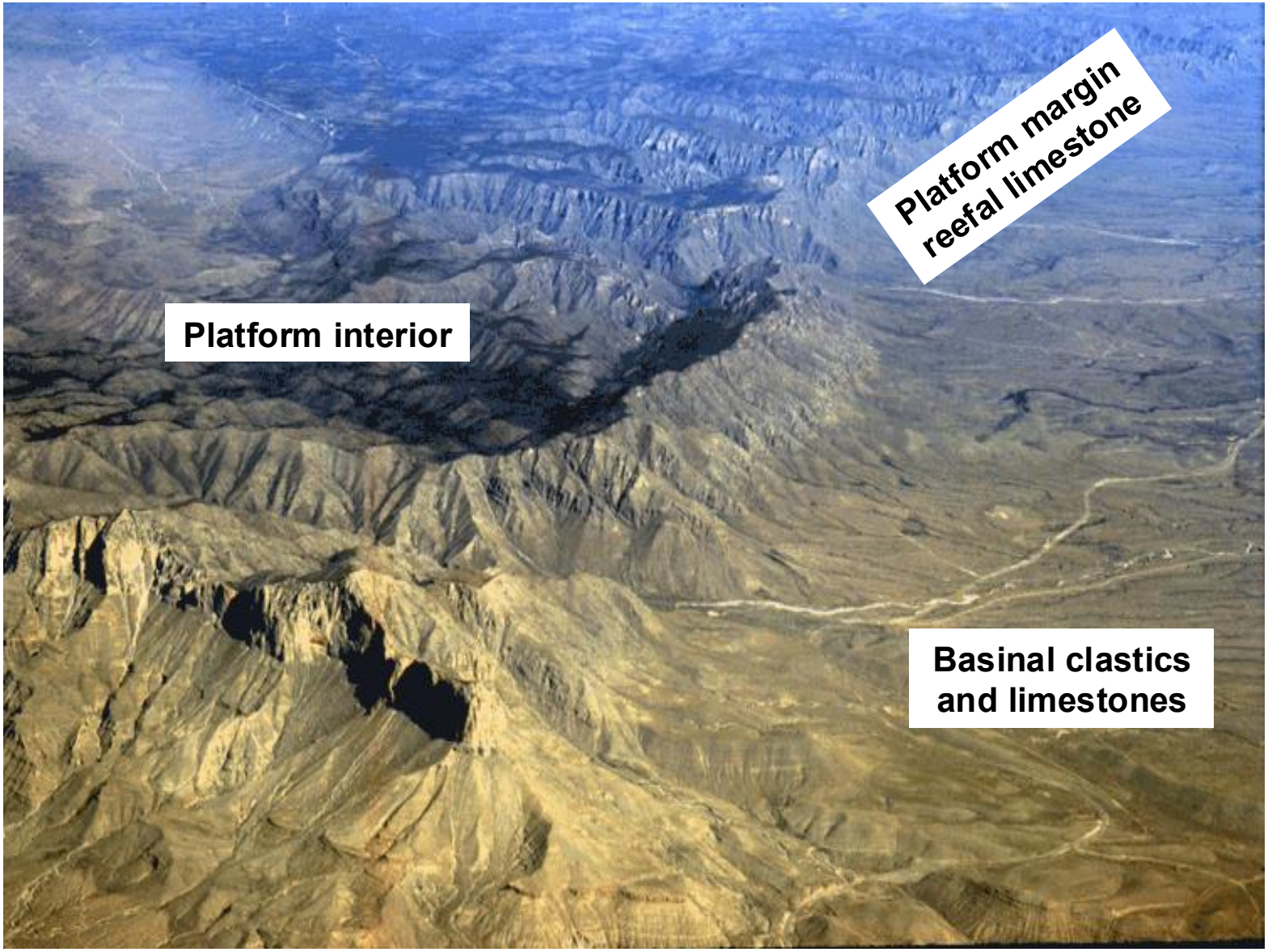
- Rimmed shelves are characterized by a shelf margin break, defined by reef growth or carbonate sandbodies
- The shelf margin is ocean-facing, high energy and turbulent
- Up-welling of ocean currents makes the shelf break a zone of high organic productivity
- Landward of the shelf margin is typically protected, characterised by lower energy/quieter water
- Sediment is transported offshore of the shelf break by sediment and gravity-driven processes

Carbonate Platforms
Rimmed shelf - Bahamas



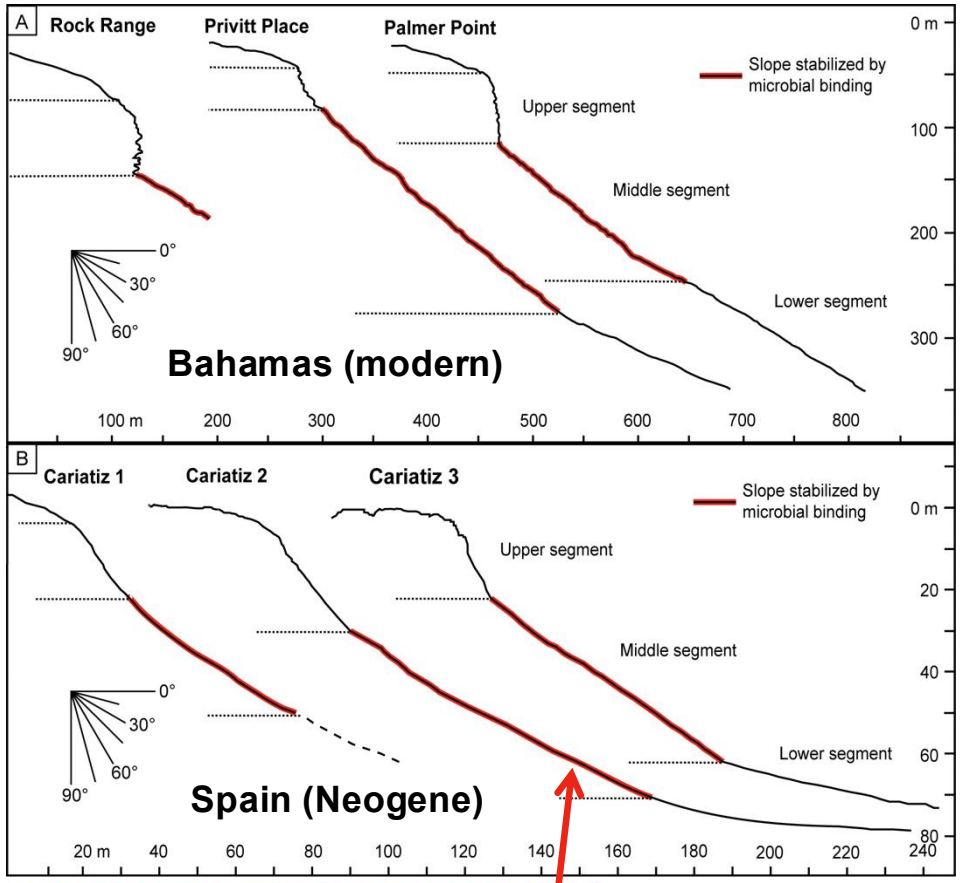
- Analog for ancient rimmed shelves (size, geometry, facies distribution)
- No clastic input
- 5,000 m of carbonate accumulation since Jurassic; <5 m Holocene sediment on karstified Pleistocene

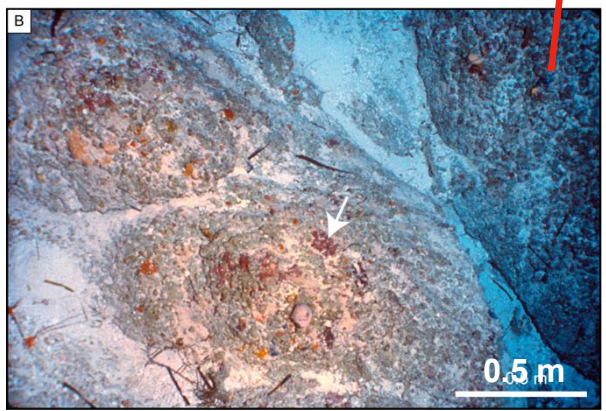
Carbonate Platforms
Rimmed shelf - Permian Delaware Basin



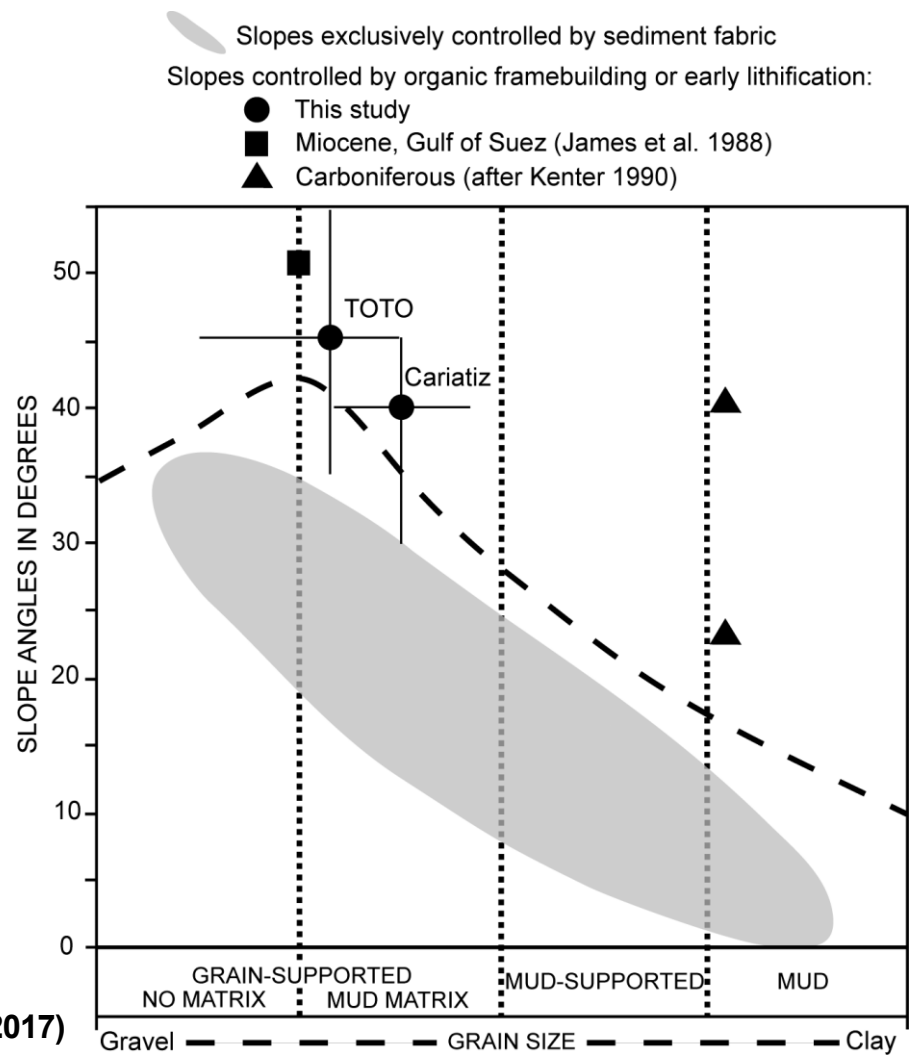
Carbonate Platforms

Platform slopes



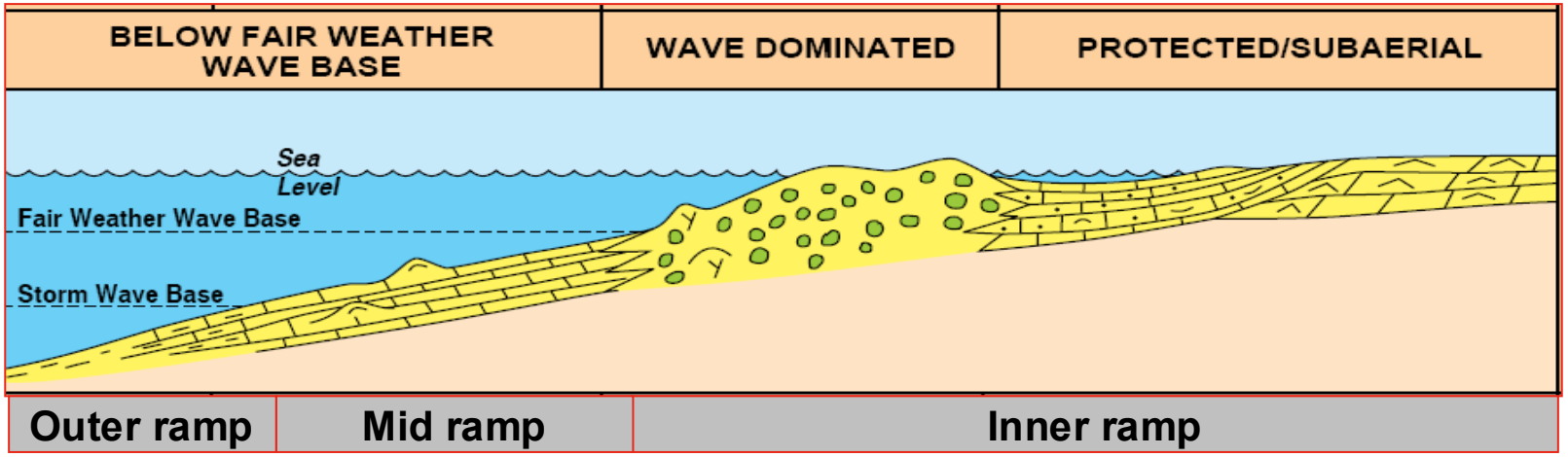


Reolid (2017)



Carbonates build steeper slopes than clastics due to early lithification and binding

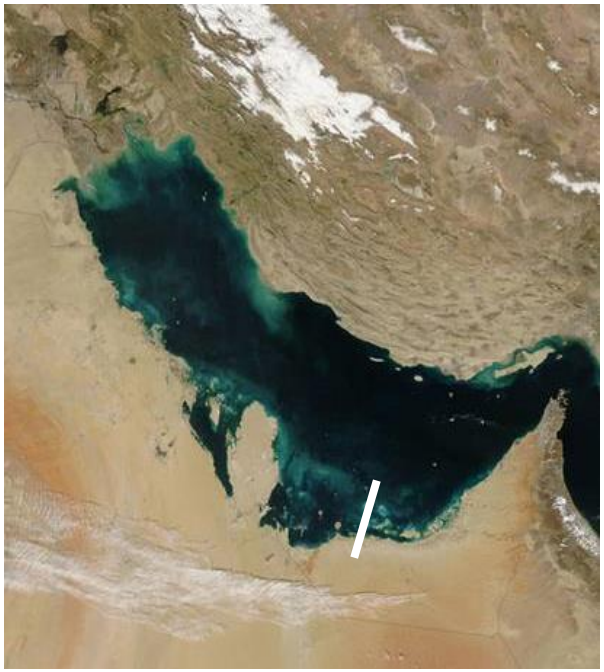
Carbonate platforms
Ramp

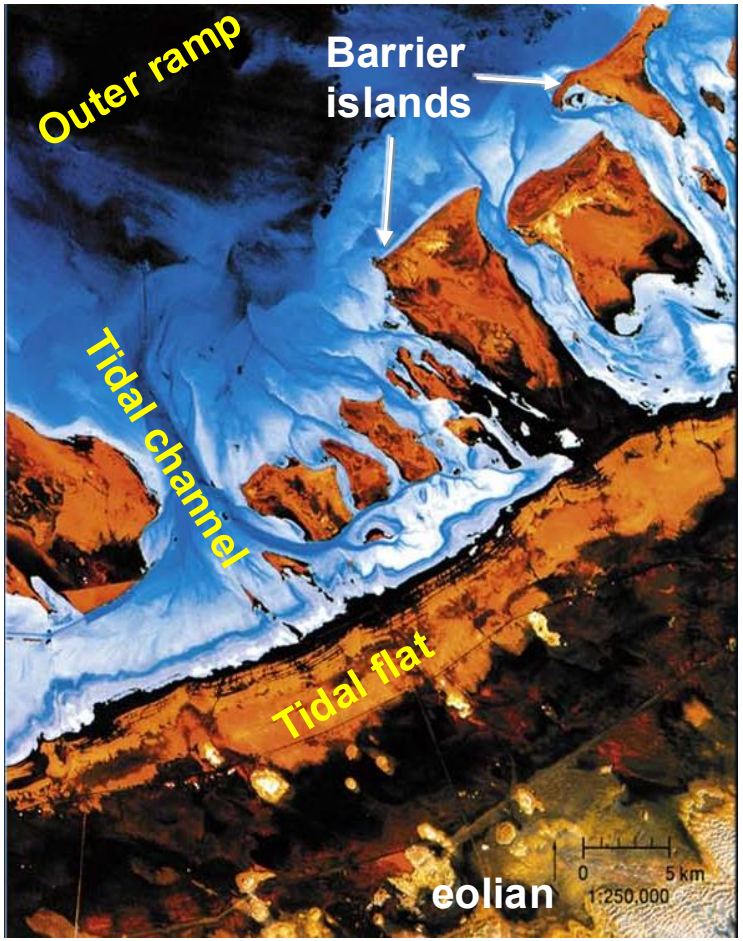


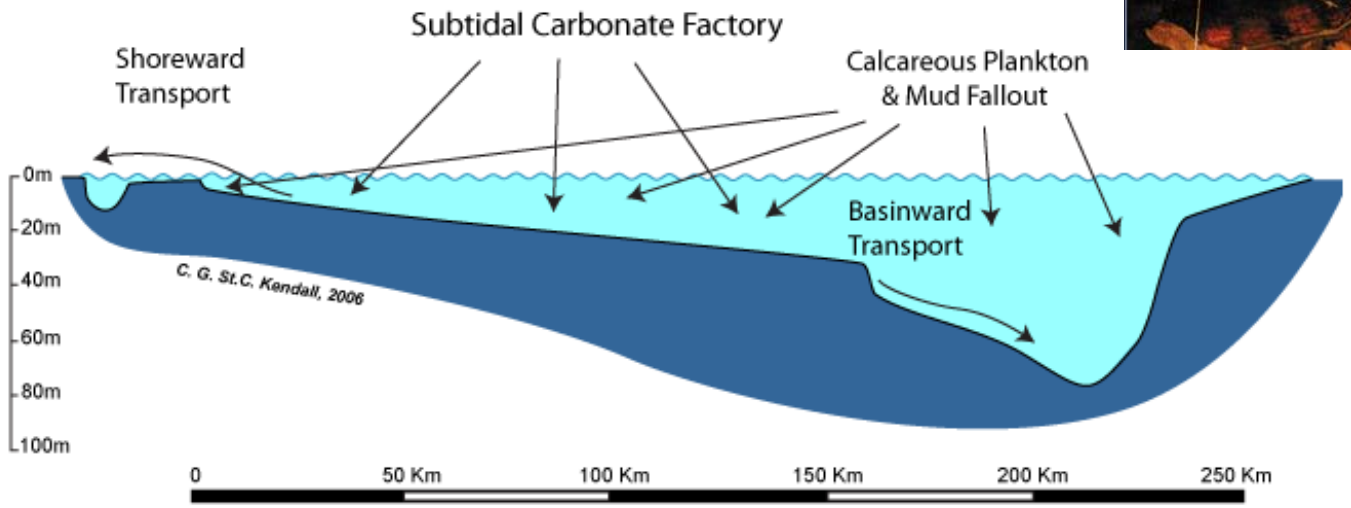
Moore, 2001

- Ramps are shallow-dipping carbonate platforms (often <1º)
- The ramp is subdivided on the basis of the influence of waves/ storms on sedimentation
- Sand shoals may develop in the high energy/wave agitated inner ramp, offering some landward protection from wave/storm energy
- Sedimentation in the mid ramp is frequently storm influenced, whilst the outer ramp is only infrequently storm-reworked
- The basin is very low energy with negligible storm influence

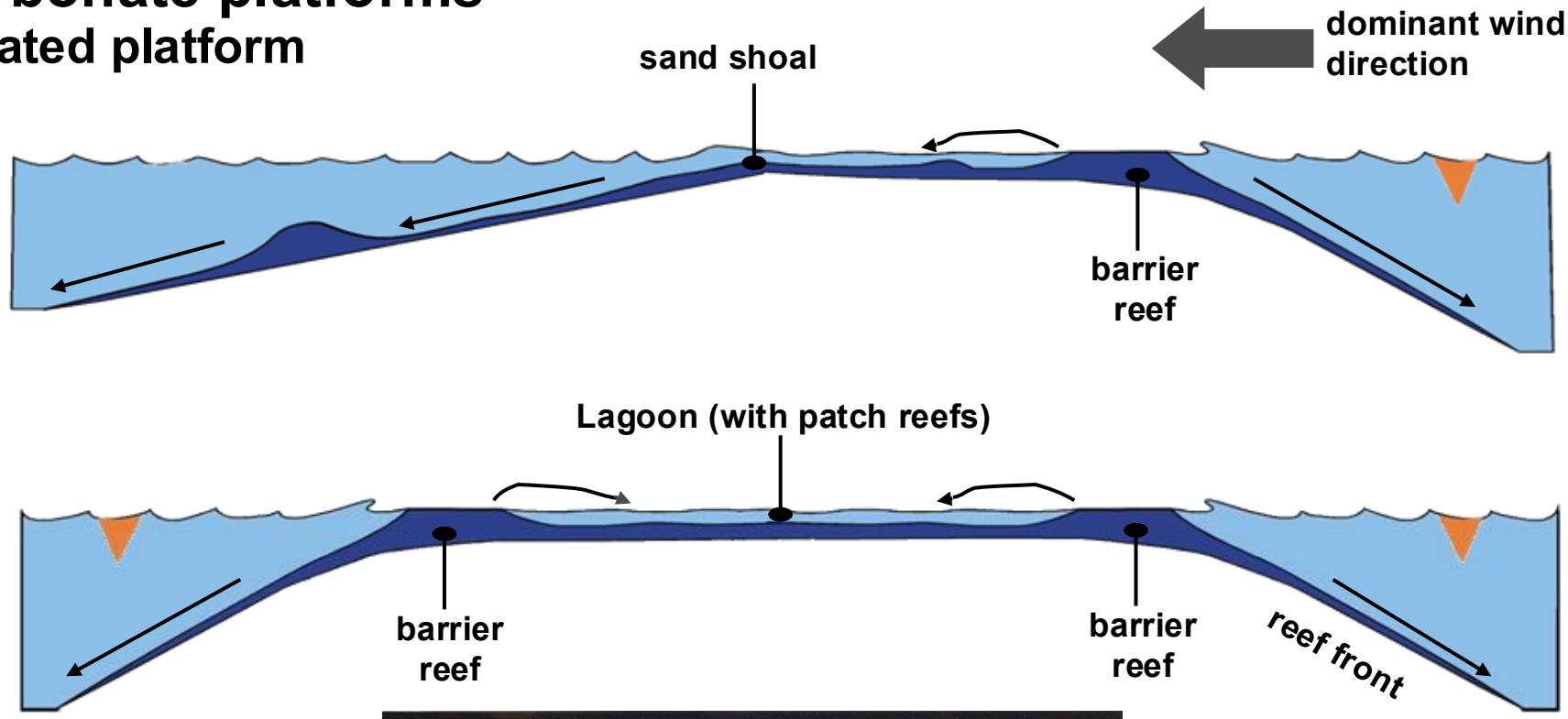
Carbonate platforms
Ramp – Arabian-Persian Gulf







Carbonate platforms
Isolated platform



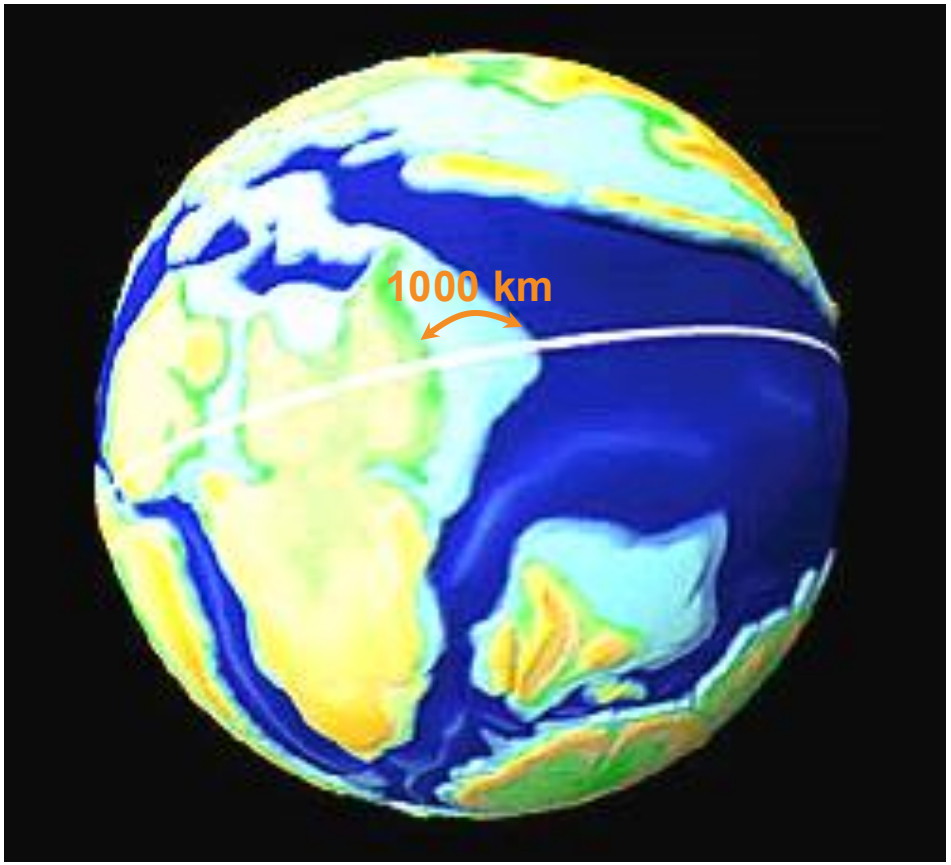
The Eye of the Maldives



Carbonate platforms
Epeiric platforms

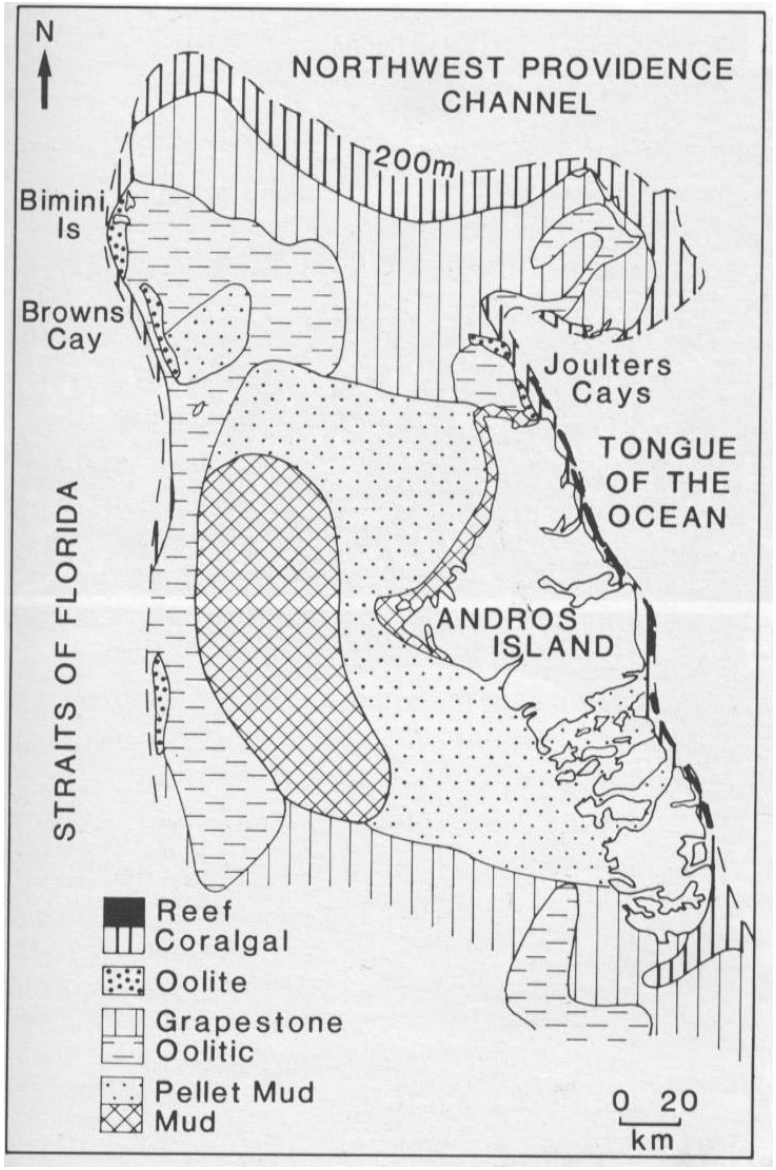
- Epeiric platforms were deposited in epi-continental seas
- Platforms < few thousands of km wide with negligible topography
- no modern analogues as present day sea level is relatively low
- Dominated by shallow water, storm-influenced, shallow subtidal-intertidal sedimentation
- Wide facies belts
- Examples:
  - Permian – Cretaceous Middle East

Mid Cretaceous (Albian)
100 million years ago



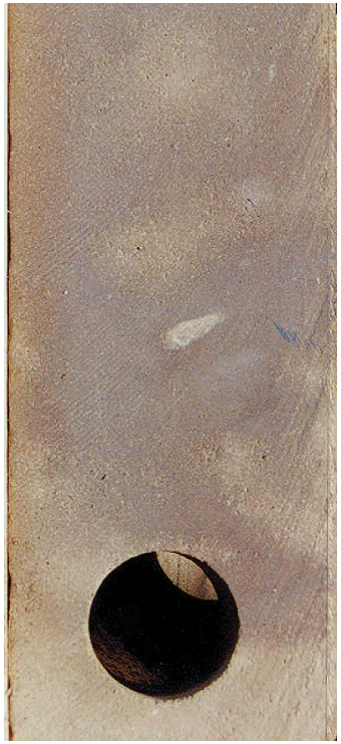
Scotese Paleomap project

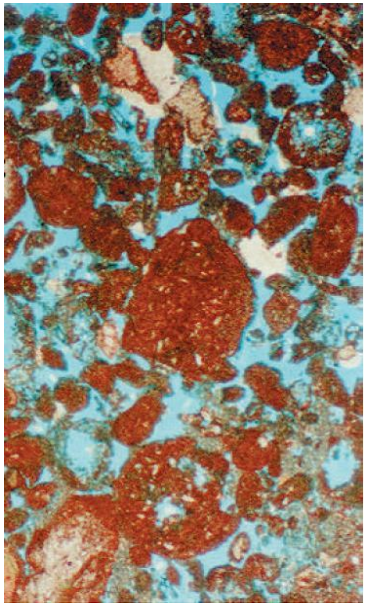
Depositional environments + lithofacies



Tucker, 1990

- Factors favouring carbonate production:
  - Warm water
  - Elevated salinity
  - Protection from terrigenous influx
- Distribution of modern sediments reflects physical environment: water depth, nutrient supply, energy
- Lithofacies: biogenic & abiogenic sediment comp.
- Facies analysis (macro- and microfacies): reconstruct depositional environments & processes





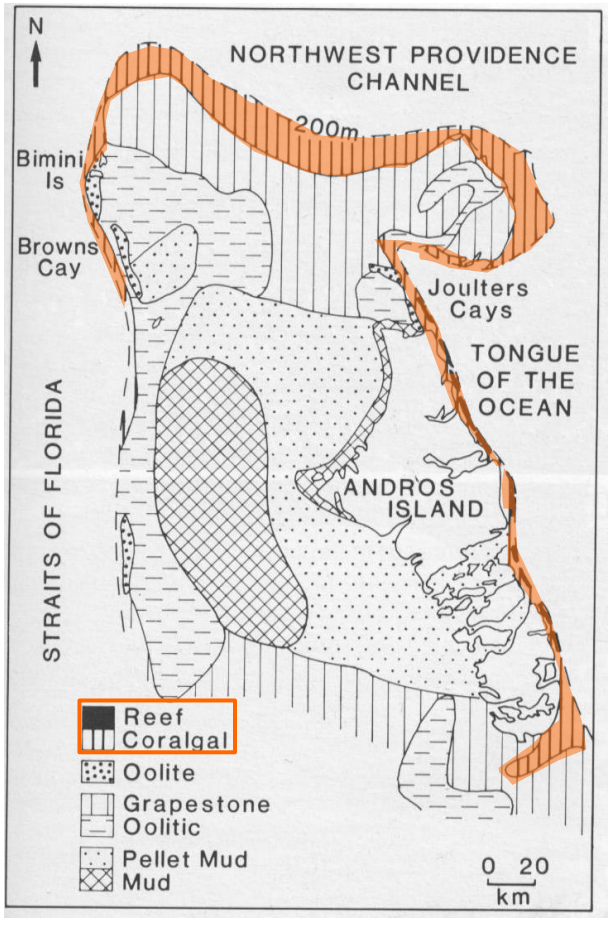
Skeletal grainstone
Algal peloidal grainstone

Depositional environments + lithofacies
High-energy platform margin: reefs

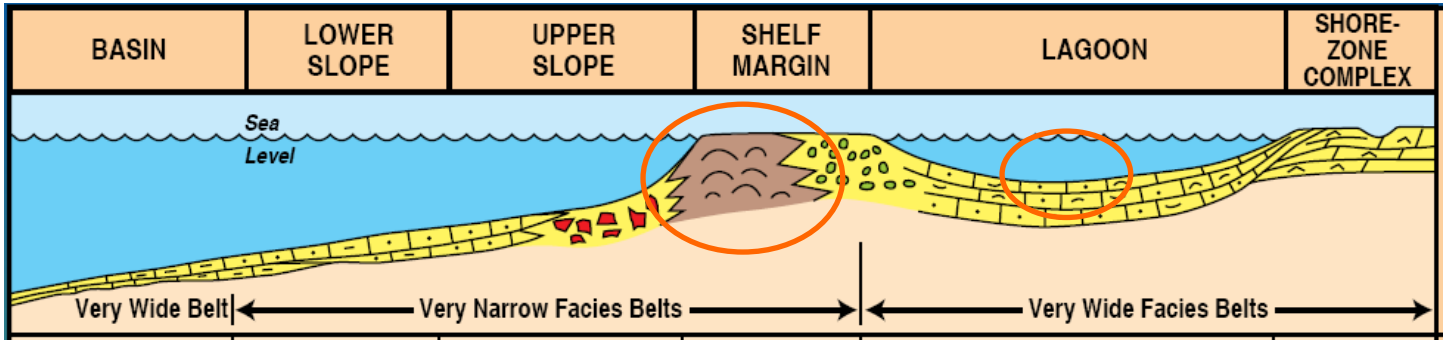




Ed Purdy/SEPM

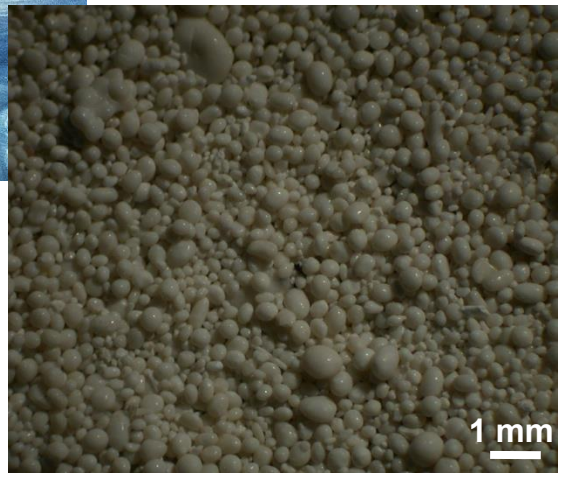


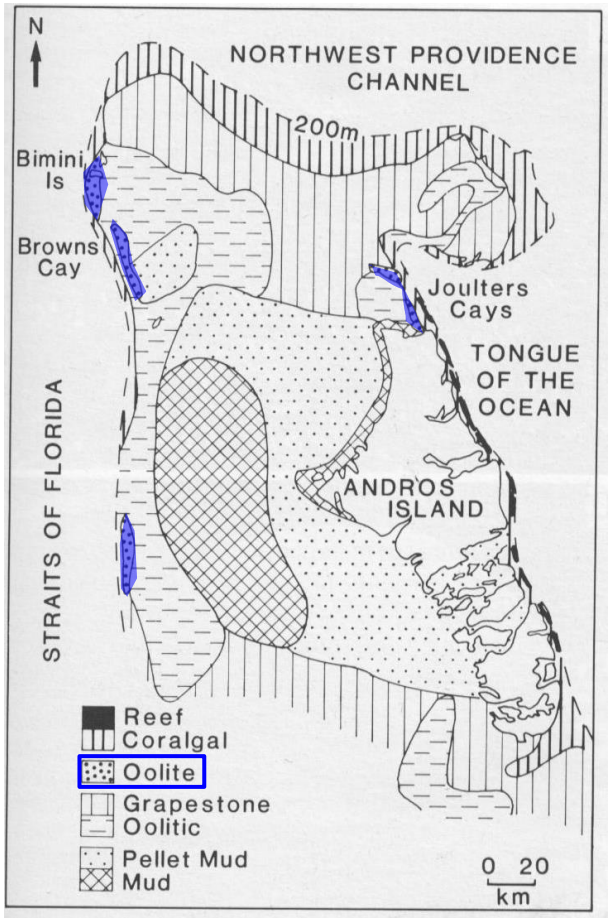
- Belt along windward margin and small patches in lagoon
- Depositional processes:
  - In-situ growth of carbonate build-ups
  - Reworking by storms, currents: rubble
  - Export to basin by gravity processes



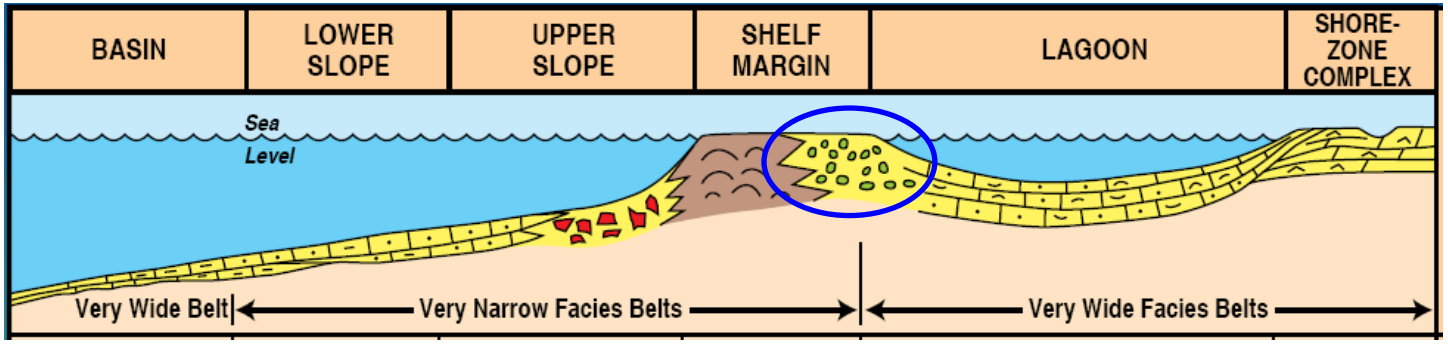
Depositional environments + lithofacies
Oolite carbonate sand shoals







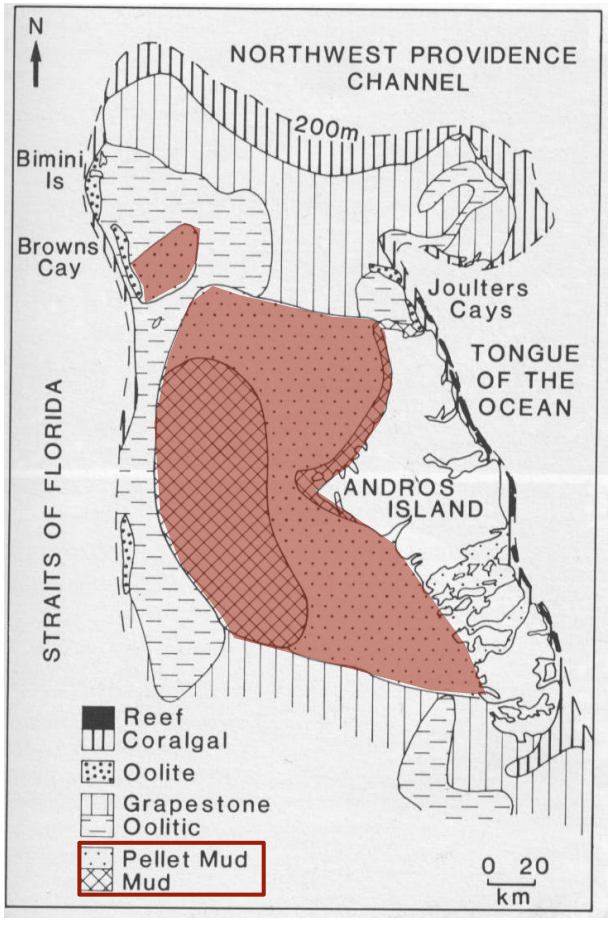
- High-energy locations
- Depositional processes:
  - In-situ precipitation of ooids
  - Reworking by tidal currents, storms, waves



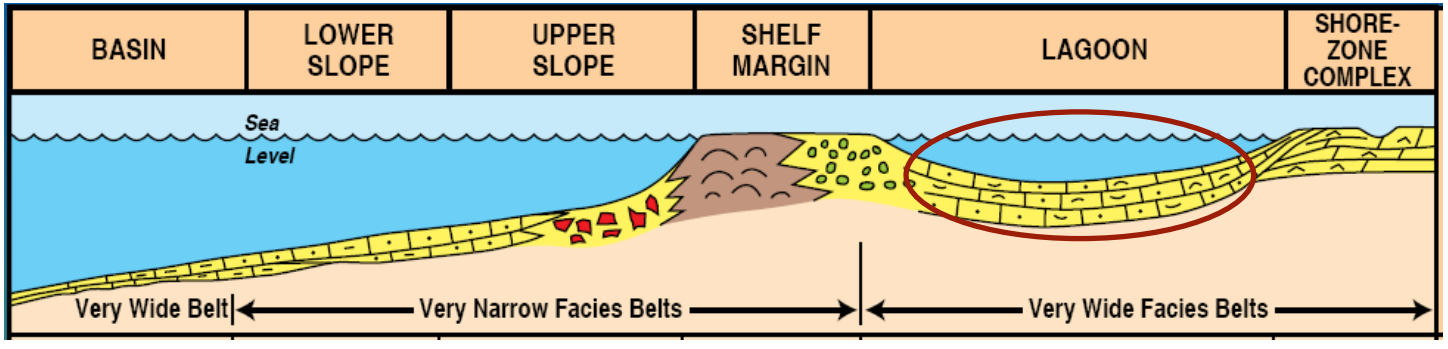
Depositional environments + lithofacies
Bank interior/lagoon







- Low-energy areas of platform
- Depositional processes:
  - Accumulation of faecal pellets (worms, gastropods)
  - Disintegration of algae: mud
  - Chemical precipitation: whitings

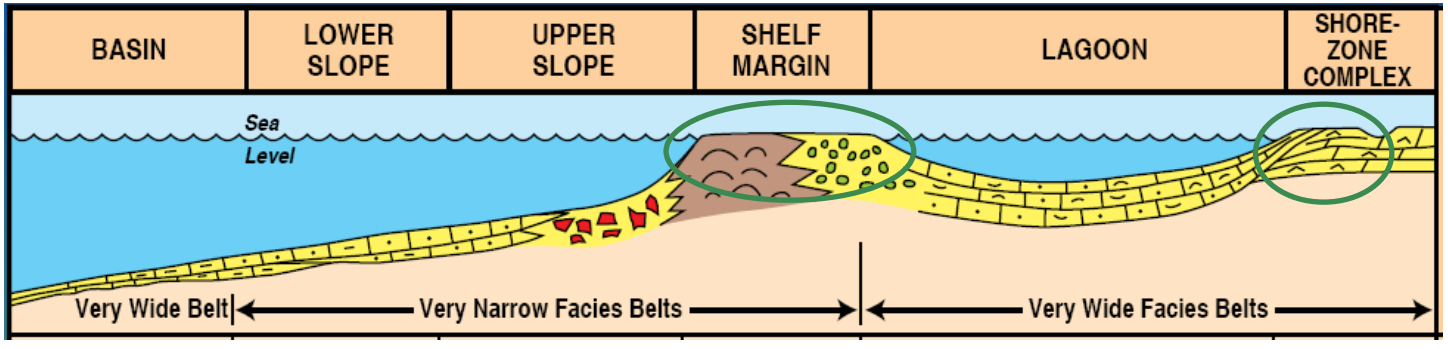


Depositional environments + lithofacies
Tidal flats

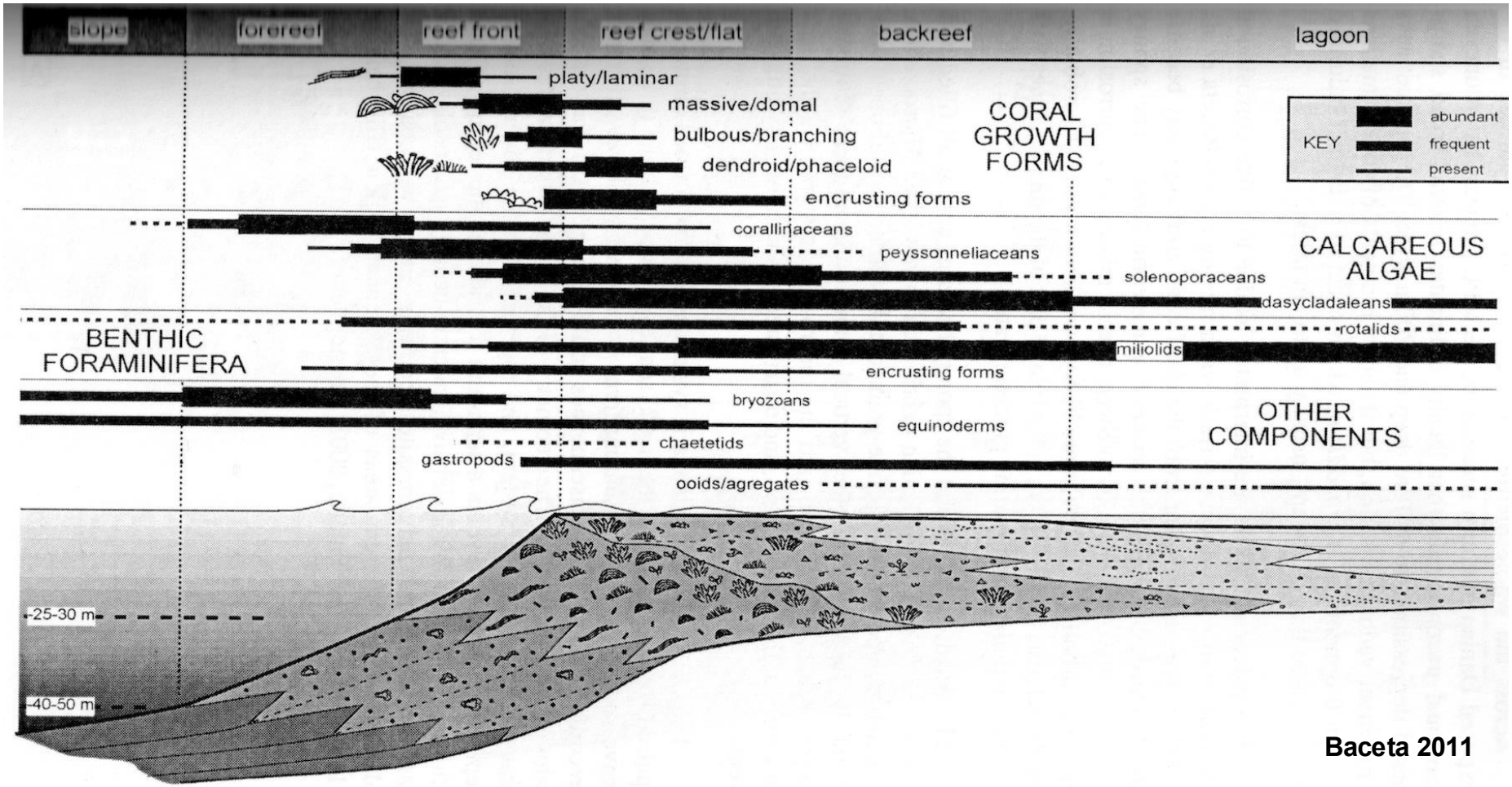




- Land-attached areas
- Depositional processes:
  - Reworking by tidal currents and waves
  - Storms move subtidal sediment onto tidal flat and into lagoon

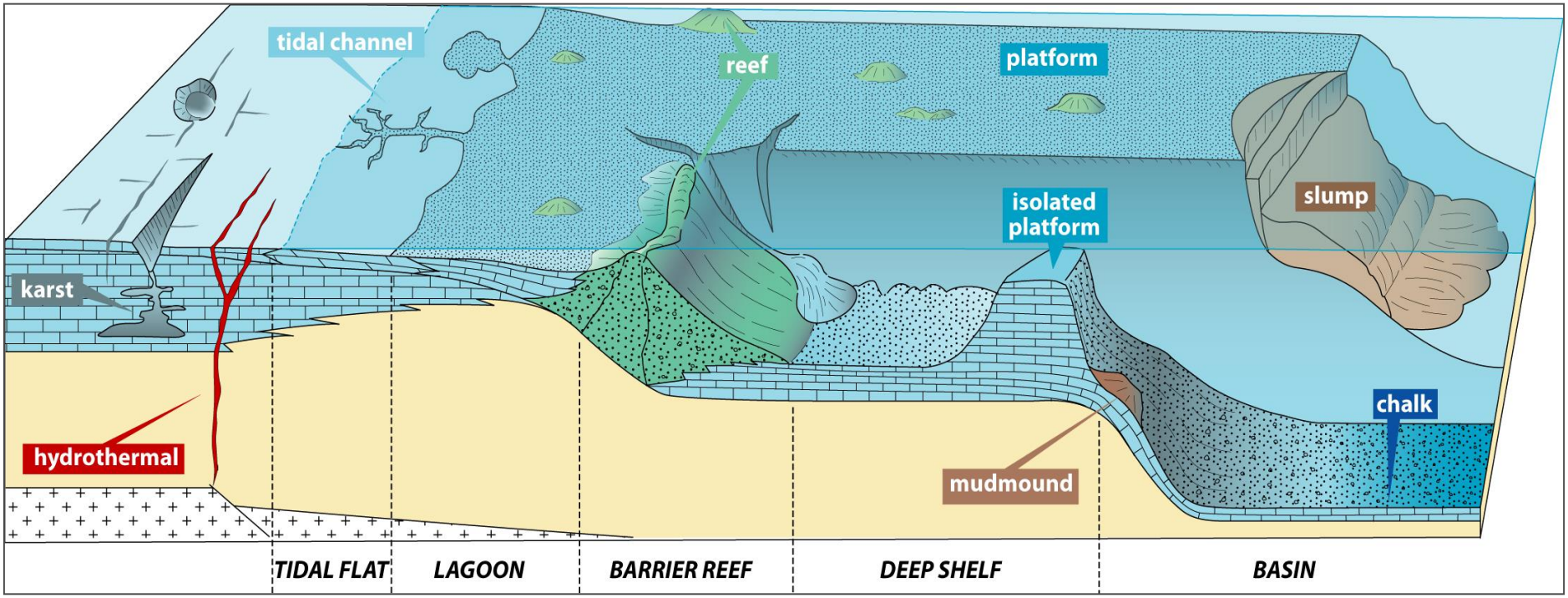


Carbonate depositional environments



Baceta 2011

Carbonate depositional environments



Lapointe 2007

Summary

- **Carbonate precipitation and accumulation are driven by carbonate factory in shallowest parts of ocean. As a consequence, carbonates build up to and track sea level.**
- **Carbonate platforms are sites of carbonate production: rimmed shelves, ramps, isolated platforms, epeiric shelves.**
- **Bahamas platform is a good example of a rimmed shelf. The distribution and character of lithofacies are controlled by characteristic environmental parameters and can be used to interpret ancient carbonate platforms.**